Introduction to Veterinary Terminology

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7.1 Introduction to Terminology for Small Animals

Learning Objectives

By the end of this chapter, you should be able to

1. Identify common anatomical landmarks in dogs and cats
2. Define sex and age-specific terms for dogs and cats
3. Describe the diversity of breeds present in dogs and cats, and how it might affect personality
4. Identify common coat colours and their terms in dogs and cats
5. Describe common husbandry procedures, forms of enrichment, and training equipment in dogs and cats
6. Define different ways of feeding and types of food given to dogs and cats
7. Describe the abbreviation AAFCO and its role in pet food

8. Discuss the diseases protected by the core vaccines in dogs and cats
9. Describe the importance of parasite prevention in dogs and cats
10. Identify some common routine and emergency causes for veterinary visits for dogs and cats

Chapter Overview

In this chapter, we will discuss terminology related to dogs and cats.

We will explore the different terms associated with their anatomy, colours, breeds, and husbandry. We will also discuss some principles of feeding dogs and cats, as well as some common reasons they visit veterinary clinics.

The purpose of this chapter is to familiarize you with the general care of cats and dogs, which will help you when booking appointments and discussing health conditions of pets with clients.

7.2 Cat and Dog Species Terms and Anatomy

Species Terms

You are certainly familiar with terms such as "cat" and "dog." However, in the veterinary field we sometimes refer to these species by an adjective relating to their scientific name.

Table showing common names and adjectives for cats and dogs

Common Name	Adjective
Cat	Feline
Dog	Canine

Within every species, we have unique names for different age and sex groups. This is especially important in production animals and breeding animals, so these terms are used more frequently for large animals than for companion animals.

We often spay or neuter our dogs and cats, but tend to just state their reproductive status rather than using a different term. Knowing reproductive status is very important, as it can help the veterinarian rule out certain diseases.

Canine

Table outlining the sex, age, and terms associated with canines

Sex	Age	Term
Intact male	Adult	Stud or dog (M)
Neutered male	Adult	Dog (M/N or MN)
Intact female	Adult	Bitch or dog (F)
Spayed female	Adult	Dog (F/S or FS)
Either	Baby	Puppy or whelp

Other terms:

- **Pack:** Group of dogs
- **Whelping:** Giving birth to a canine

Feline

Table outlining the sex, age, and terms associated with felines

Sex	Age	Term
Intact male	Adult	Cat (M)
Neutered male	Adult	Cat (M/N, MN)
Intact female	Adult	Queen (F)
Spayed female	Adult	Cat (F/S, FS)
Either	Baby	Kitten

Other terms:

- **Queening:** Giving birth to a feline
- **Feral:** a domestic cat that has lived outdoors for an extended period of time and is not socialized to humans.

Exercises



An interactive H5P element has been excluded from this version of the text. You can view it online here:
https://pressbooks.openeducationalberta.ca/veterinarymedicalterminology/?p=2017#h5p-71

Basic Anatomy

Anatomical terms are very important to help describe the location of injuries, so it is crucial to have a general knowledge of these **topographical** markers.

The general layout of the skeleton of cats and dogs is very similar to the human skeleton. The table below lists anatomical terms for cats and dogs, along with their human counterpart.

From Head to Toe

The following table discusses some of the common anatomy landmarks you will hear in companion animal medicine.

Table of anatomical terms and definitions

Anatomical Term	Definition
Muzzle	Equivalent to human nose and lips The most rostral portion of the head
Pinna	Ear flap
Shoulder	Equivalent to human shoulder Joint where the scapula (shoulder blade) meets the humerus (front limb)
Elbow	Equivalent to human elbow Joint where the humerus meets the radius and ulna (front limb)
Carpus	Equivalent to the human wrist Joint between the radius/ulna and the metacarpals
Dewclaw	Residual toe that does not bear weight between the carpus and the foot. Often present on front limbs, occasionally on hindlimbs

Paw	Equivalent to human foot The foot of the animal that they walk on
Stifle	Equivalent to human knee Joint between the femur and the tibia/fibula (hind limb)
Hock/tarsus	Joint equivalent to human ankle Tarsus is the name of the joint, but it is colloquially referred to as the "hock" in most animals (hind limb)
Tail	Anatomy differs depending on species, but in general caudal extension of spine +/- extra hair

Comparative Anatomy – Dog vs. Human

Animals and humans share the same general skeletal layout. In Chapter 8, there are figures showing a horse and a human skeleton, colour-coded to show how they compare, and a comparison between the hock in the cat and the horse.

Below, we will show how the front leg in the dog relates to the arm of a human.

Note: Phalanges are the bones that make up our fingers and toes!

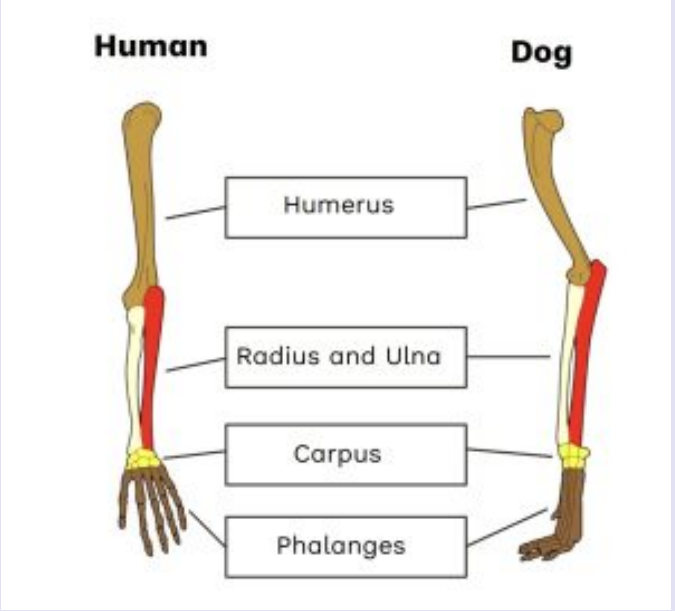


Figure 7.1 Comparative anatomy

Dogs

The following image shows the locations of the terms discussed above on a picture of a dog.

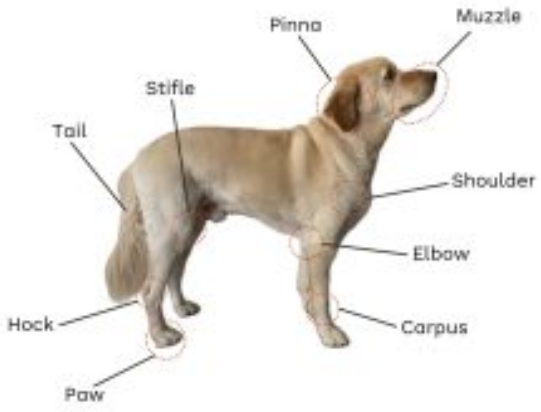


Figure 7.2 Canine anatomy

Cats

The following image shows the locations of the terms discussed above on a picture of a cat.

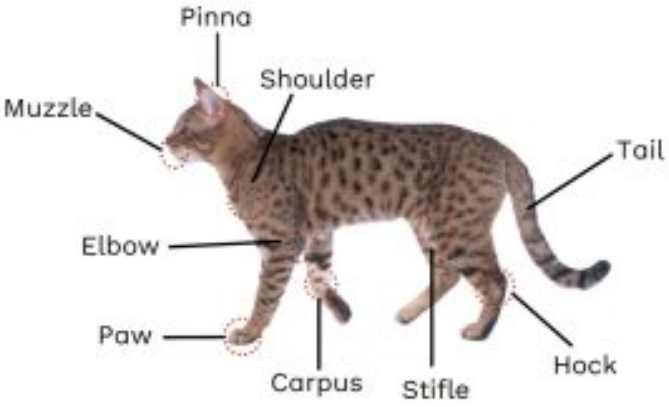


Figure 7.3 Feline anatomy

Exercises

An interactive H5P element has been excluded from this version of the text. You can view it online here:

7.3 Cat and Dog Breeds and Colours

Breeds

Dogs and cats are very popular pets in Canada—they live in approximately 60% of Canadian households (Canadian Animal Health Institute, 2022). Rover, a popular network for pet boarding, found that the most popular breeds in Canada were mixed breeds, followed by the Labrador retriever and the golden retriever (Rover, 2023a). In a review of the most common cat breeds in Canada, they found the domestic shorthair and domestic longhair to be quite popular. These are the equivalent of "mixed breeds" in the cat world, and most people find the colour of the cat to be more important than the breed (Rover, 2023b). From this author's experience in the clinic, you are much more likely to encounter purebred dogs than purebred cats.



Figure 7.4 Labrador retriever (black)



Figure 7.5 Golden retriever



Figure 7.6 Mixed breed dog



Figure 7.7 Domestic longhair cat



Figure 7.8 Domestic shorthair cat

Clinical Insight

The terms domestic short, medium and long hair are used in felines to describe unknown or mixed breed and length of hair coat. The acronyms are: DSH, DMH, & DLH.

There are associations in many countries that serve as registries for **purebred** animals. A purebred animal is one who is a single breed, generally recognized by a registry, with no other breeds

present in its heritage. In Canada, the principal purebred dog registry is the **Canadian Kennel Club (CKC)**, and they determine the breed standards for every purebred in Canada (Canadian Kennel Club [CKC], n.d.). The cat equivalent in Canada is called the **Canadian Cat Association (CCA)** (Canadian Cat Association [CCA], n.d.a).

Breed is an important factor to consider in the clinic. Different breeds of dogs and cats can be more likely to develop certain diseases, and certain breed and colour combinations have risks as well. For example, white cats with blue eyes are more likely to be deaf.

Different breeds of dogs were bred for different jobs. For example, the standard poodle was used as a water retriever for duck hunting (American Kennel Club [AKC], n.d.a), while Siberian huskies were bred for sled pulling in freezing temperatures (AKC, n.d.b), and the shih tzu was raised as a lap dog for Chinese emperors (AKC, n.d.c).



Figure 7.9 Shih tzu



Figure 7.10 Standard poodle



Figure 7.11 Siberian husky

As you might imagine, and have probably experienced, the breed of a dog can have an effect on personality, implying a genetic relationship. A recent study suggested that socialization as a puppy could also affect personality moving forward (Salonen et al., 2023).

Despite the frequent association of some breeds with friendliness, it is important to remember that even within a breed each individual animal can differ from the norm. Therefore, it is important not to make assumptions in the clinic. It is always advisable to listen to the owner when they tell you about their dog or cat, as they know them best.

Note: In some breeds it is common to **tail or ear dock**, which means surgically remove portions of the tail or ears. This is a

contested procedure and legality varies by province when done for cosmetic purposes. The Alberta Veterinary Medical Association (ABVMA) has banned its members, veterinarians, and **RVT**s from doing these procedures if not medically necessary (Canadian Kennel Club, 2019).

Additional Common Breeds

Here are some photos of different dog breeds to further demonstrate the diversity within the species.



Figure 7.12 Irish wolfhound



Figure 7.13 Greyhound



Figure 7.14 English bulldog



Figure 7.15 Rottweiler

Colours

Purebred dogs and cats have standardized acceptable colours, as noted by their registries (CKC, n.d., CCA, n.d.a).

Common Canine Coat Colours

- **Black**
- **Chocolate**
    - Brown
- **Fawn**
    - Beige, often with a black face and muzzle
- **White**
- **Merle**
    - Generally white coat with another colour spots (red vs. blue)
- **Brindle**
    - Generally black and orange striped coat
- **Tri-Colour**
    - Combination of brown, white, and black



Figure 7.17 Coat colour: chocolate



Figure 7.16 Coat colour: black

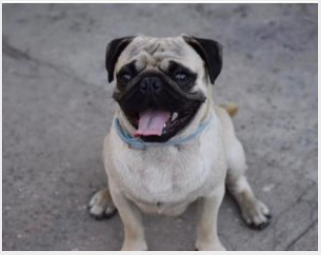


Figure 7.18 Coat colour: fawn



Figure 7.19 Coat colour: white



Figure 7.20 Coat colour: blue merle



Figure 7.22 Coat colour: brindle



Figure 7.21 Coat colour: tri-colour

***Note*:** There are many more colour patterns and solid colours seen in dogs, these are just a sample of some common ones you will encounter.

Common Feline Coat Colours

- **Tortoiseshell (tortie)**
  - Black and red coat
    - "Red" is the term used by the CAA. Normally this is called "orange" in clinic.
- **Calico**
  - White, black, and red coat
    - "Red" is the term used by the CAA. Normally this is called "orange" in clinic.
- **Tabby**
  - Colour pattern often involving splotches and stripes
- **Points**
  - Refers to the face, paws, ears, and tail
- **Solid colour**

- Cat is entirely one colour



Figure 7.23 Coat colour: grey tabby



Figure 7.24 Coat colour: tortoiseshell



Figure 7.25 Coat colour: brown tabby



Figure 7.26 Coat colour: calico

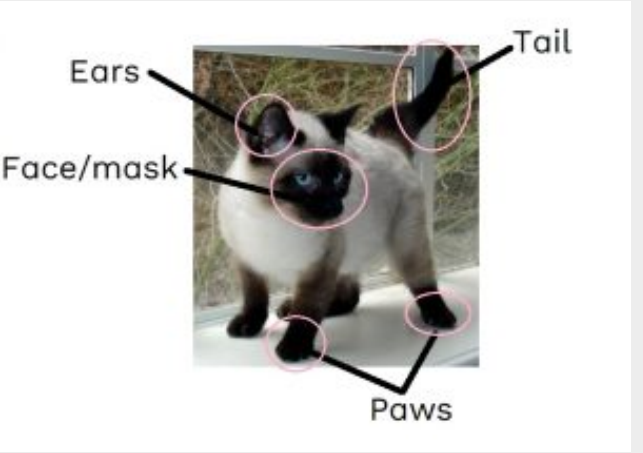


Figure 7.27 Location of "points" on a cat

Note: There are many more colour patterns and solid colours seen in the cat world, these are just some common terms you will encounter.

Colour descriptions adapted from CCA colour and pattern handouts (CCA, n.d.b).

7.4 Cat and Dog Husbandry

This section is of the book explores some common pet care principles and terms for dogs and cats, including socialization, grooming, exercise and enrichment, equipment training, and litterboxes.

Socialization

When puppies and kittens are young, they should be **socialized.** This means exposing them to different stimuli, including surfaces, people, children, and noises, to help get them used to different environments. Socialization helps them to be less anxious should they encounter these stimuli later in life. Getting them used to handling is also very important; vet visits are much less stressful for pets if they are used to people handling them. Socialization must have a balance though, as we do not want to frighten, stress, or over-stimulate a puppy or kitten (Root Kustritz, 2022).

Grooming

Both dogs and cats require some form of regular grooming. What each requires will depend on the coat characteristics. Some dogs, like poodles and shih tzus, do not shed very much, but their hair continuously grows, so it needs to be cut. Other dogs continuously or seasonally shed their coats, so it always remains approximately the same length. Though these dogs might not get haircuts, they still require regular bathing and brushing to prevent **matting**.



Figure 7.28 Cat grooming itself

Cats often self-groom by licking themselves with their barbed tongue. However, they cannot always reach every part of their body, so they might require some extra brushing. Long-haired breeds in particular benefit from regular brushing.



Figure 7.29 Cat's tongue grooming hair

Another common procedure is nail trims. Both dogs and cats need their nails trimmed because, like humans, their nails continuously grow. See integumentary system for more information.

Groomers: People whose profession it is to groom dogs. This often includes bathing, brushing, haircuts, and nail trims.



Figure 7.30 Cat getting his nails trimmed



Figure 7.31 Grooming tools. From left to right: brush, de-matting tool, comb, nail trimmers

Exercise and Enrichment

Exercise and enrichment are important to both cats and dogs. This might include walks outside, playing fetch in the park, or doing agility classes. However, there is much more we can do to enrich the lives of our pets than exercise.

Puzzle feeders: One way to increase mental stimulation is to make dogs and cats work for their food. Puzzle feeders make pets solve puzzles to access their food, which is mentally stimulating for them.

Lick mats: Wet food, mushy bananas, plain yogurt, or peanut butter can be smeared on a mat that is textured, so an animal will take longer to lick it up. These foods can also be frozen to make play time last longer. Lick mats make feeding time last longer while increasing satisfaction.

Toys and play: Animals can play individually, with their owners, or with other animals. Toys can be a good enrichment resource, though care must be taken if there are multiple animals in the home, as this can result in resource guarding. The play must be supervised to prevent a pet form eating what they should just be playing with.

Cat trees: These are posts with platforms and hide-aways for cats that allow them to get up high.



Figure 7.32 Sample of dog toys



Figure 7.33 Cat playing with a wand toy



Figure 7.35 Sample of puzzle feeders and a lick mat



Figure 7.34 Two cats in cat tree

Training Equipment – Dogs

- **Rewards:** Examples of rewards include food, play, and affection. Different dogs are motivated by different rewards, so it is important to know which works best for the dog (Root Kustritz, 2022).
- **Clickers:** A tool used in clicker training, which is a form of training where the dog is taught to associate a "click" sound with a food reward. The handler clicks when the desired behaviour occurs, then gives the dog a treat. Timing is important–the click marks the behaviour and the treat comes after the click (Root Kustritz, 2022).
- **Halters:** Head halters include the Gentle Leader and Halti collars. Both allow control of the dog's entire head. The Gentle Leader permits the handler to gently close the dog's mouth and reduces pulling, barking, mouthing, and jumping (Root Kustritz, 2022).
- **Buckle collars:** Collars that wrap around the neck and have a quick-release clip or an actual buckle–most commonly used type of collar (Root Kustritz, 2022).
- **Harnesses:** These fit around the chest of the dog and clip in the front or on the back, which makes pulling by the dog less efficient. These give handlers more control over the dog. These should be removed when not in use (Root Kustritz, 2022).
- **Leashes:** Leashes that are 4–6 feet in length are recommended for training. They may be made of nylon or leather.
    - Retractable leads are often used, but do not offer much control. The lock can fail, and fingers can be severely injured if they become entangled in the leash cord (Root Kustritz, 2022).
- **Crates and kennels:** Crates and kennels can be used as safe-places within a living environment, as a sleeping space, or for travel. Dogs should be crate-trained if they are going to be spending time in one so that they can be comfortable in that

space (Root Kustritz, 2022).

- **Gates:** Gates can be made of plastic, metal, or wood. They are great for multi-dog households and households with dogs and children. They can be used to prevent dogs from dashing through open doors, but must be properly installed or they can be knocked over, and of an appropriate height or they can be jumped over (Root Kustritz, 2022).

***Note*:** Many of these can also be used with cats, but they are not as common.

Cats – Litterboxes

Just like with food, cats can be picky about their litterboxes. There are many different styles; some are covered and some are uncovered, the height of the rim varies, as does size. There are also many different types of litter with different textures, scents, and materials. A rule of thumb to determine how many litter boxes a household should have is number of cats plus one. So if you have one cat, you should have two boxes and if you have four cats, you should have five boxes. Boxes should be placed away from where food and water are provided, in a semi-private location away from disturbances and high traffic, in an area that is somewhat open so the cat does not feel cornered or trapped (Root Kustritz, 2022).

Feces and clumped urine should be scooped out at least once daily. All litter should be dumped out, the box completely cleaned with warm water and mild detergent, and new litter placed in the box every two weeks (Root Kustritz, 2022).

Clinical Insight

Many clients are concerned their cats do not use the litter box properly out of anger or spite, but this is not the case. Generally there is an underlying medical or behavioural problem, or the litter box doesn't meet the cat's preferences (Root Kustritz, 2022).



Figure 7.36 Example of a cat litterbox filled with litter

7.5 Cat and Dog Nutrition

Feeding

Dogs and cats are fed by life stage. Dietary goals include good quality of life, maximum longevity, optimization of performance, and disease prevention. Obesity is the number one nutritional disease in dogs and cats, and is present in 12–40% of animals. Health risks such as musculoskeletal disease, cardiovascular disease, cancer, diabetes mellitus, reduced immunocompetence, and decreased life span are all associated with obesity. Other conditions associated with nutrition are dental disease and **urolithiasis** (Root Kustritz, 2022a).

Determination of food dosage is the basis for nutritional management, and is a basic skill in nutritional therapy. We must calculate food dosage with the understanding that dogs and cats may be fed a complete and balanced diet, supplements, and treats (Root Kustritz, 2022a).

Specific diets are produced to feed dogs and cats for specific life stages or activities. Examples include diets for growth, lactation, adults at maintenance, athletes / performance, and geriatric animals. Some diets are breed-specific. Veterinarians can prescribe diets to treat specific diseases; highly digestible diets may be prescribed for intestinal disorders, and diets with unique proteins may be required for dogs that are allergic to common pet food ingredients (Root Kustritz, 2022a).

We often rely on a nutritional adequacy statement from the **Association of American Feed Control Officials (AAFCO)** to determine if a pet food is nutritionally balanced. While AAFCO does not regulate pet food, they provide guidelines for the nutritional

needs of different species of pets at different life stages such as growth, gestation, lactation, maintenance, or all of the above. Dog and cat foods will have labels on them describing that they meet AAFCO guidelines for a specific species and life stage (Rishniw, 2023).

Clinical Insight

While both species can be picky, cats often have specific preferences for temperature, odour, and physical form of food (Root Kustritz, 2022a).

Types of Food

- **Kibble**
  - Dry food, bite-size chunks
- **Canned**
  - Wet food in a can, generally classified as paté (smooth, blended consistency) or stew (chunky)
- **Treats**
  - Not nutritionally balanced, high palatability, meant for intermittent feeding
  - Includes dental sticks (dogs), pill pockets, and chews (dogs)
- **Home-cooked**
  - Owners making their pet's meals with human food

    - Note: These should follow a recipe or be formulated by a veterinary nutritionist to ensure they are properly nutritionally balanced
- **Raw**
    - These can be home-made or commercial, and are composed of raw meat, bones, and some vegetables
    - Risk of nutritional imbalance and bacterial contamination
- **Grain-free (dogs)**
    - Can be wet or dry, contain a high proportion of peas, lentils, other legumes, and other ingredients including potatoes
    - Potentially linked to heart disease in dogs, but not confirmed yet
- **Vegetarian**
    - Often does not contain enough proteins for cats, who are carnivores

(Root Kustritz, 2022a)



Figure 7.37 Examples of dog food, from left to right: Treats, dehydrated raw food, Greenies (dental stick – back), canned food (paté), kibble



Figure 7.38 Cat with bowl of canned food. Bag of kibble behind cat.

Mixing Medications with Food

Some medication can be mixed with food, which makes it easier to administer. A veterinarian should confirm if medications can be mixed together and whether or not to give them with food. Some brands create treats that are

specifically designed to hold pills, to make oral administration easier. Medications can sometimes also be mixed with canned food so the pet has a harder time picking out the medication from the food. Pets also find the flavour more appealing.



Figure 7.39 Pill Pockets for dogs: Treats that are formulated to be soft and hold pills



Figure 7.40 Pill Pockets for cats: Treats that are formulated to be soft and hold pills



Figure 7.41 This picture shows canned food with a paste medication and powder medication about to be mixed together.

Ways of Feeding

Free choice feeding = ad libitum or ad lib.

- Food is freely available at all times
- Increased risk for obesity and decreased awareness of how much a pet is eating
- Only for dry food, as wet food, raw food, and home-cooked

food will go bad

Food restricted or meal feeding

- Measured amount of food is given at intervals throughout the day
- Best control and knowledge of intake, and best control of body weight

(Root Kustritz, 2022a)

Body Condition Score

Body condition scoring: System of assigning a numerical score based on physical characteristics indicative of fatness. These include the amount of muscle and fat covering different areas of the body (Kustritz, 2022b).

When determining the body condition score of dogs and cats, we often use a **scale of 1–9**. Large animals are done on a **scale of 1–5**. Therefore, it is important to always mark what the score is out of to avoid confusion, as 4/9 is ideal while 4/5 is overweight.

Generally, on a 9-point scale, 4 or 5 are ideal body weight while 3 is ideal on a 5-point scale.

How To: Body Condition Score

The following links give guides on how to body condition score a dog and a cat on a 9 point scale. They will show you what to look for and where on the body we see notable changes.

- Dog BCS Guide (9-point scale)
- Cat BCS Guide (9-point scale)

It is important to actually palpate the animal when doing body condition scoring. This is especially important when it is difficult to visualize the areas of interest, such as in fluffy dogs.

Exercises

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7.6 Cat and Dog Vet Visits

Cats and dogs might routinely visit the veterinary clinic in order to be vaccinated, have a wellness exam, and get parasite prevention. They also come to the clinic when they are experiencing health problems. This section will give an overview of some wellness visits you might encounter, as well as a couple common emergencies.

Vaccination

The purposes and benefits of vaccination are outlined in Chapter 6. Each species has its own recommended vaccine protocols that vary depending on geographic region and risk of exposure (Kustritz, 2022a).

In North America, many clinics follow the vaccinating guidelines for dogs and cats set out by the **American Animal Hospital Association (AAHA)** (Ellis et al., 2022; Stone et al., 2020). Generally, puppies and kittens start their first vaccine series at 6–8 weeks of age and are vaccinated every 3–4 weeks until they are about 16 weeks of age (Ellis et al., 2022; Stone et al., 2020). This can vary depending on circumstance, and it is important you refer to your clinic's guidelines for when vaccines should occur. Puppies and kittens require a revaccination about 1 year after completion of the initial vaccine series to develop protection against disease (Kustritz, 2022b). Required frequency of rabies vaccination is regulated by province in Canada. In some provinces, it is legally required for dogs and cats to have up-to-date rabies vaccines (e.g. Ontario), while others it is not (e.g. Alberta).

Core vaccinations are the sets of vaccinations that are suggested for all dogs and cats regardless of where they live or what activities

they do. The additional vaccines, termed non-core, are given depending on the lifestyle of the pet and the risks they encounter (Kustritz, 2022c).

Common Core Vaccines

Canine Diseases

- **Canine distemper (D)**
  - Viral disease that causes a respiratory infection along with neurological changes
  - Highly contagious to other dogs, can be lethal
- **Canine adenovirus or hepatitis (A_2 or H)**
  - Viral disease that causes liver disease
  - Contagious to other dogs, can be lethal
- **Canine parvovirus (P)**
  - Viral disease that causes severe vomiting and bloody diarrhea
  - Highly contagious to other dogs, can be lethal
- **Rabies (R)**
  - Viral disease that causes neurological signs and is ultimately fatal
  - Contagious to all mammals, including

humans

(Root Kustritz, 2022b)

Feline Diseases

- **Feline panleukopenia (P)**
  - Viral disease causing severe vomiting and diarrhea
  - Contagious to other cats
- **Feline rhinotracheitis or herpesvirus (FVR or FHV)**
  - Viral disease causing upper respiratory disease and **ocular** discharge
  - Contagious to other cats
- **Feline calicivirus (C)**
  - Viral disease that causes respiratory disease
  - Contagious to other cats
- **Rabies (R)**
  - Viral disease that causes neurological signs and is ultimately fatal
  - Contagious to all mammals, including humans

(Root Kustritz, 2022b)

Vaccines often combine multiple viruses into one vaccine. For example, the **DAP** or **DHP** vaccine contains three core canine vaccines: ***distemper, adenovirus/hepatitis, and parvovirus***. The feline equivalent core vaccine is **RCP** or **FRCP** standing for ***rhinotracheitis, calicivirus and panleukopenia.*** The "F" stands for feline.

Clinical Insight

A common combination vaccine for dogs is **DAPP,** also called **DA2PP**. Like **DAP,** this vaccine protects against distemper, adenovirus/hepatitis, and parvovirus, as well as parainfluenza. It is commonly given via the SQ route (under the skin).

Most injectable vaccines for dogs and cats are given under the skin. The kennel cough vaccine is often given orally or intra-nasally.

Common Non-Core Vaccines

Canine Diseases

- **Kennel Cough**
    - Disease characterized by a harsh cough. Often caused by the bacteria Bordetella bronchiseptica (Bord.) or the parainfluenza virus
    - Highly contagious, but generally animals recover quickly
- **Leptospirosis (L)**
    - Bacterial disease causing **renal** and hepatic disease
    - Contagious to other animals and humans

(Root Kustritz, 2022b)

Feline Diseases

- **Feline leukemia (FeLV)**
    - Viral disease that causes elevated **leukocytes** and immunosuppression
    - Contagious to other cats
    - may be considered a core vaccination in

shelter cats and/or cats less than one year old

(Root Kustritz, 2022b)



Figure 7.42 Oral kennel cough vaccine ready for administration

Clinical Insight

Daycares and boarding for dogs and cats often have specific vaccination requirements. It is important that the owners are aware of what these are before coming to the clinic, so they can tell the veterinary team what their pet requires for daycare or boarding. It is also important to remember that oftentimes these centres will have a mandatory wait-time between vaccine administration and hosting pets, as it takes about a week for the vaccine to fully take effect.

Parasite Control

In dogs and cats, we generally try to prevent external parasites like lice, fleas, and ticks, and regularly treat for internal parasites like intestinal worms. Deworming for internal parasites is often a crucial part of the first few veterinary visits, because puppies and kittens tend to get worms from their dams. Different intestinal parasites require specific deworming medication, so we often run fecal tests to determine which internal parasite is present and which treatment is required (Root Kustritz, 2022d).

Another type of internal parasite we protect against is heartworm. This parasite is transmitted by mosquitoes and lives in the bloodstream, eventually going to the heart. Due to the difficulty in

treating this parasite, we often aim to prevent it, rather than treat it (Root Kustritz, 2022d).

External parasites can cause intense skin irritation and itchiness, but they can also transmit diseases. For example, the deer tick can transmit Lyme disease, and fleas can transmit some intestinal worms (Root Kustritz, 2022d).

Many different anti-parasite drugs exist and they cover a varied scope of parasites. These drugs can be **topical,** meaning they are applied to the skin, or **oral,** meaning they are given by mouth. The frequency with which they are given depends on the drug, the season, and the patient. The choice of drug is very patient-specific for what they require protection against, their lifestyle, and where they live (Root Kustritz, 2022d).



Figure 7.43 Bravecto box: Oral flea and tick preventative for dogs >20–40 kg.

Clinical Insight

Some external parasite products made for dogs are toxic to cats! Therefore, it is important for a veterinarian to know if there are other pets in the household so they can properly advise on how to mitigate risk, or choose a different product.

Routine Visits

- **Annual exam**
  - Most pets are recommended to visit the vet yearly for check-ups and to maintain a valid veterinary-client-patient-relationship.
  - This check-up allows owners to discuss wellness procedures like flea or tick control and vaccines, and increases the probability that diseases will be caught early.
- **Spay/Neuter**
  - Surgeries to remove gonads in male and female dogs and cats are very common in North America.
  - Females – ovaries and uterus removed
  - Males – testes removed
  - Age at which this is best done depends on health conditions, breed/size, and lifestyle. It should be discussed with a veterinarian.

- **Anal gland expression**
  - Anal glands and sacs are paired and located on either side of the anus. They are supposed to secrete their content as fecal matter pushes against them, but not all pets are able to do this properly. Some pets come in for regular anal gland expression, which means to empty them.
- **Dentistry**
  - Can be provided via routine teeth cleaning, which usually requires general anaesthesia
  - X-rays of the mouth are often taken to further visualize dental disease that is not visible during an oral exam
  - Further procedures might include extractions

Euthanasia

Euthanasia is when animals undergo a planned, humane death. This is an extremely difficult time for clients, whether it has been planned for a week or occurs in an emergency. No amount of preparation makes this easy, and it is crucial to be compassionate in these times. Many clinics will have protocols to follow for euthanasias, including when the clients pay (before vs. after vs. at pick-up of ashes), and which rooms are used.

General process:

- Pets will often undergo an exam by a veterinarian to determine if euthanasia is warranted.

- Clients must sign a consent form before euthanasia. In Canada, this form will include a guarantee that the animal has not bitten anyone in a certain time period. This is a rabies safety precaution.
- Many veterinarians will use sedation to calm the pet before placing an IV catheter, which is used to administer an overdose of an anesthetic. This causes the animal to die within a few minutes.
    - There are other ways euthanasia is done, but this is one of the most common.
- Death must be confirmed by the veterinarian, often by confirming there is no heartbeat, respiration, or corneal reflex.
- Owners can elect to take the bodies home and bury them (each municipality has regulations on how this should be done), or they can be cremated. In this author's experience, most owners elect to cremate their pets.
    - Different cremation options exist depending on which crematorium is used. There can be communal cremation where the pet is cremated with other animals; ashes may or may not be returned, and if they are, they are a mix of all the pets. Cremations may also be private, where the pet is cremated alone, and generally the ashes are returned to the owner. Many pet crematoriums offer memorial items such as ink paw prints and nose prints, clay paw prints, and hair clippings.
    - Oftentimes, pet crematoriums pick up bodies

from the clinic, and sometimes pick-up at clients' houses can be arranged.

Common Emergencies

- **Hit by car (HBC)**
  - Animal has been hit by a car
  - Can lead to broken bones, hemothorax, and other life-threatening conditions
  - Treatment depends on what is going on, but diagnostics to determine extent of damage are important
- **Blocked cat**
  - Cat that is not peeing, or only dripping urine
  - Most common in male cats
  - This is an emergency, because the bladder can grow and burst if urine cannot be released.
  - Treatment might include passing a urinary catheter to unblock the cat, and supportive therapy moving forward. Surgery is sometimes considered.
- **Foreign body**
  - Animal has eaten something they should not have that can cause a blockage in the digestive tract
  - Treatments might include inducing vomiting, depending on what the object is and how long ago it was consumed, and exploratory laparotomy to remove the object
- **Toxicities**
  - It is common for dogs and cats to eat things that are toxic to them, especially around holidays
  - These might cause neurological, kidney, and/or

gastrointestinal issues, among others
- Severity depends on what was eaten, and in what quantity
- Treatments might include inducing vomiting, giving activated charcoal, gastric lavage or supportive therapy
- Some examples include:
    - Food like chocolate, coffee, tea, grapes, onions, garlic
    - Plants such as lilies for cats
    - Cleaning products
    - Antifreeze
    - Pesticides
    - Rat poison



Figure 7.44 Easter lilies are extremely toxic to cats



Figure 7.45 Chocolate is toxic to dogs and cats

Animal Poison Control Centre

The **ASPCA (American Society for the Prevention of Cruelty to Animals)** runs a poison control centre that both pet owners and veterinary teams can contact regarding potential toxicities (ASPCA, n.d.).